Limits and Continuity

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July 24, 2025

1 Introduction

To 'compile" this information I am using the book Calculus by David Patrick it can be found here: `https://artofproblemsolving.com/store/book/calculus`.

This Document is designed to review the Limit and Continuity principals covered in the aformentioned book. I will create other "Guides" for other topics covered. They can be found here: `https://github.com/KovasMcCann/Writings`.

2 Limits

2.1 Definition

A **limit** describes what happens as a function "approaches" a point in $\mathbb{R}$. Syntacticly it is represented as:

$$\lim_{x \to a} f(x) = L$$

Where the values of $f(x)$ get arbitrarily close to L as x gets arbitrarily close to a, from **both sides**.

2.2 Input and Output

Understanding limits requires a clear grasp of input-output relationships. Recall that:

- The **domain** is the set of all possible inputs (x-values), denoted $\text{Dom}(f)$.
- The **range** is the set of all possible outputs (y-values), denoted $\text{Rng}(f)$.

Keep in mind that its all **possible** inputs and **not** all inputs. Functions can also be visualized on the Cartesian plane, where each point $(x, f(x))$ represents an input-output pair. Thinking this way helps clarify how functions behave near critical points or discontinuities.

2.3 δ–ϵ Definition of a Limit

Let f be a real-valued function. We say that the limit of $f(x)$ as x approaches a is L, written as

$$\lim_{x \to a} f(x) = L,$$

if, for every $\epsilon > 0$, there exists a $\delta > 0$ such that whenever $0 < |x - a| < \delta$, it follows that $|f(x) - L| < \epsilon$. That is,

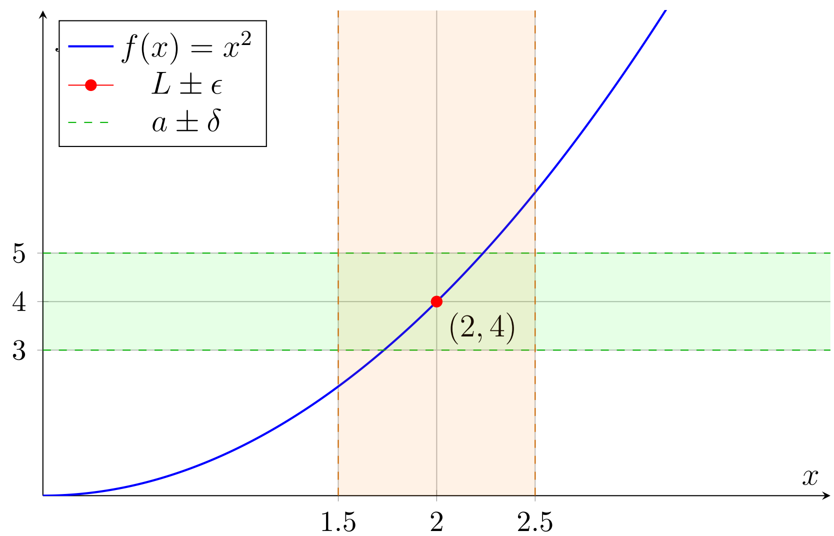
$$\forall \epsilon > 0, \ \exists \delta > 0 \text{ such that } 0 < |x - a| < \delta \Rightarrow |f(x) - L| < \epsilon.$$

Graphical Representation

Let $f(x) = x^2$, and consider the limit

$$\lim_{x \to 2} f(x) = 4.$$

Given an arbitrary $\epsilon > 0$, such as $\epsilon = 1$, we want to find a $\delta > 0$ (here, choose $\delta = 0.5$) such that whenever $0 < |x - 2| < \delta$, it follows that $|f(x) - 4| < \epsilon$. The graph below shows this visually:



Key

- The blue curve shows $f(x) = x^2$.
- The red point marks $(2, 4)$, where the limit is taken.
- The green dashed lines represent the ϵ-band: $y = 4 \pm 1$, i.e., $(3, 5)$.
- The orange dashed lines mark $x = 2 \pm 0.5$, the δ-interval: $(1.5, 2.5)$.
- The shaded regions visualize the intervals within which x and $f(x)$ must lie.

As shown, for all x in the interval $(2 - \delta, 2 + \delta)$, the corresponding values of $f(x)$ lie within $(L - \epsilon, L + \epsilon)$, confirming the δ–ϵ condition.

2.4 Explination

Epsilon and delta help define the approach of a point that you measure. Or more formaly they establish a precise relationship between the input and output of a function, ensuring continuity in their behavior near a point.

2.5 Uniqueness

Limits must be unique, meaning any given limit must approach a single, consistent value. For example take two limits

$$\lim_{x \to a} f(x) = L_1, \quad \lim_{x \to a} f(x) = L_2$$

where $L_1, L_2 \in \mathbb{R}$. Let $\varepsilon > 0$ be arbitrary.

By the definition of the limit, there exist $\delta_1, \delta_2 > 0$ such that:

$$\begin{aligned} 0 < |x - c| < \delta_1 \text{ and } x \in A &\implies |f(x) - L_1| < \frac{\varepsilon}{2}, \\ 0 < |x - c| < \delta_2 \text{ and } x \in A &\implies |f(x) - L_2| < \frac{\varepsilon}{2}. \end{aligned}$$

Let $\delta = \min(\delta_1, \delta_2) > 0$. Since c is an accumulation point of A, there exists $x \in A$ such that $0 < |x - c| < \delta$.

Then both conditions apply, and we have:

$$|L_1 - L_2| = |L_1 - f(x) + f(x) - L_2| \leq |L_1 - f(x)| + |f(x) - L_2| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, it follows that $|L_1 - L_2| = 0$, and hence:

$$L_1 = L_2.$$

This relates to the value of continuity and the idea that limits "respect" the input-output relationship of a function. As we stated above, because the input is the same, and because limits preserve a consistent connection between input and output, the outputs (i.e., the limits) must also be the same. That means any difference represented by epsilon must cancel out, leading us to conclude that the two limits are equal.

2.6 Squeeze Theorm

The Squeeze Theorem is used to create a limit of a function indirectly.

Squeeze Theorem: Let $a \in \mathbb{R}$ and leg f, g, h be real-valued functions such that $f(x) \leq g(x) \leq h(x)$ for all x in an open interval containing a (but not necessarily at a itself). If

$$\lim_{x \to a} f(x) = \lim_{x \to a} h(x) = L$$

then $\lim_{x \to a} g(x) = L$ as well.

The most important part is the inequalites because they allow to set all values equal to eachother and thus "squeezing" them

2.7 Left, Right, and Infinite Limits

A limit examines the behavior of $f(x)$ as x gets arbitrarily close to a point a. There are three types:

1. **Left-hand limit:** $\lim\limits_{x \to a^-} f(x) = L$ Means $f(x)$ approaches L as x approaches a *from values less than* a (i.e. from the left).

2. **Right-hand limit:** $\lim\limits_{x \to a^+} f(x) = L$ Means $f(x)$ approaches L as x approaches a *from values greater than* a (i.e. from the right).

3. **Infinite limits:** $\lim\limits_{x \to a^\pm} f(x) = \pm\infty$ Means $f(x)$ grows without bound (positively or negatively) as x gets close to a. This describes vertical asymptotes.

Two-sided limit exists $\iff$ both one-sided limits exist and are equal:

$$\lim_{x \to a} f(x) = L \quad \text{if and only if} \quad \lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = L.$$

If either one differs, the two-sided limit does not exist.

Infinite one-sided limits do not count as a finite two-sided limit: Even if both sides go to $+\infty$, the two-sided limit is still "does not exist" in a finite sense. We describe it as an unbounded divergence.

2.8 Continuity

A function $f(x)$ is said to be **continuous at a point** a if it satisfies all three of the following conditions:

1. $f(a)$ is defined.

2. $\lim_{x \to a} f(x)$ exists.

3. $\lim_{x \to a} f(x) = f(a)$.

If any of these conditions fail, f is **discontinuous** at a.

Continuity on an interval

- f is continuous on an interval if it is continuous at every point in that interval.
- Intuitively, continuous functions can be drawn without lifting your pen.

Types of discontinuities

Common discontinuities include:

- **Removable:** $\lim_{x \to a} f(x)$ exists but $f(a)$ is undefined or differs.
- **Jump:** One-sided limits exist but differ.
- **Infinite:** One-sided limit(s) go to $\pm\infty$.

Examples

$$f(x) = \begin{cases} \dfrac{\sin x}{x}, & x \neq 0, \\ 1, & x = 0 \end{cases} \quad \Rightarrow \quad \lim_{x \to 0} f(x) = 1 = f(0),$$

so f is continuous at 0.

$$g(x) = \begin{cases} 2x, & x < 6, \\ x - 1, & x \geq 6 \end{cases}$$

Here, $\lim_{x \to 6^-} g(x) = 12 \neq 5 = \lim_{x \to 6^+} g(x)$, so g is discontinuous at $x = 6$.

Properties of continuous functions

- Polynomials are continuous everywhere; rational functions are continuous on their domains.
- Sum, product, quotient (denominator nonzero), and composition of continuous functions remain continuous.

The Intermediate Value Theorem (IVT)

If f is continuous on $[a, b]$ and k is between $f(a)$ and $f(b)$, then there exists at least one $c \in [a, b]$ such that $f(c) = k$.

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Summary:

$$\lim_{x \to a} f(x) = f(a) \quad \text{iff} \quad \begin{cases} f(a) \text{ exists,} \\ \lim_{x \to a} f(x) \text{ exists,} \\ \lim_{x \to a} f(x) = f(a). \end{cases}$$

This ensures there is no "break" in the graph at $x = a$, enabling powerful results like the IVT and smooth functional behavior.